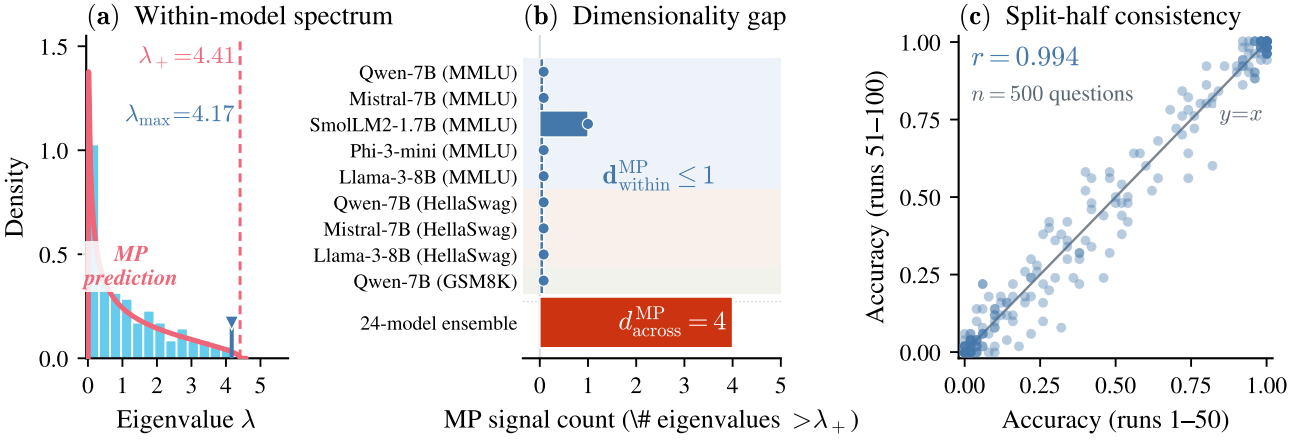

(a) Within-model spectrum
$\lambda_+ = 4.41$
$\lambda_{\max} = 4.17$
MP prediction
Density
Eigenvalue λ
(b) Dimensionality gap
Qwen-7B (MMLU)
Mistral-7B (MMLU)
SmolLM2-1.7B (MMLU)
Phi-3-mini (MMLU)
Llama-3-8B (MMLU)
Qwen-7B (HellaSwag)
Mistral-7B (HellaSwag)
Llama-3-8B (HellaSwag)
Qwen-7B (GSM8K)
24-model ensemble
$\mathbf{d}^{\mathrm{MP}}_{\mathrm{within}} \leq 1$
$d^{\mathrm{MP}}_{\mathrm{across}} = 4$
MP signal count (\# eigenvalues $> \lambda_+$)
(c) Split-half consistency
$r = 0.994$
$n = 500$ questions
$y{=}x$
Accuracy (runs 51–100)
Accuracy (runs 1–50)